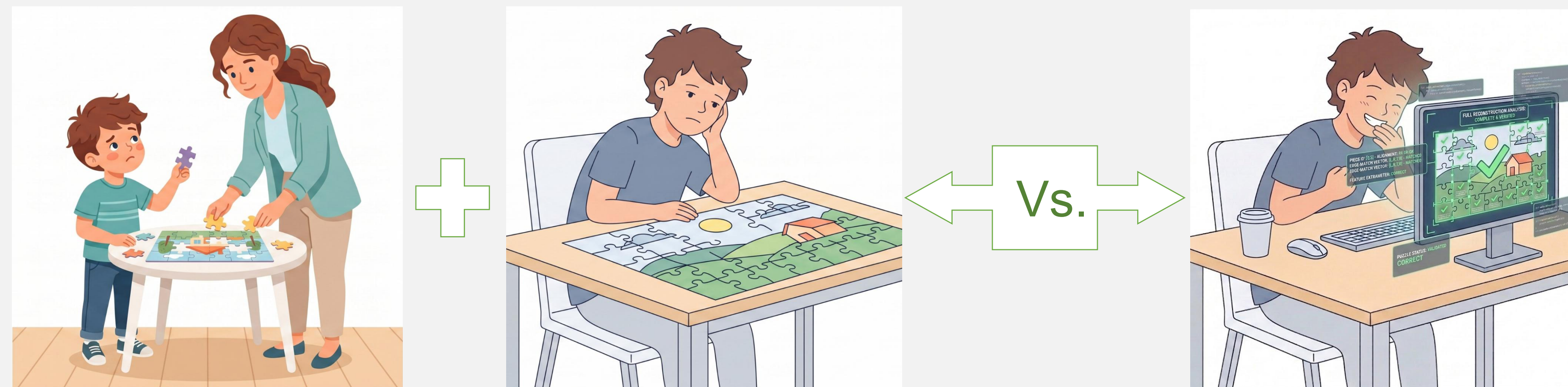


CompleteMe - Interactive Educational Puzzle Platform Utilizing Computer Vision

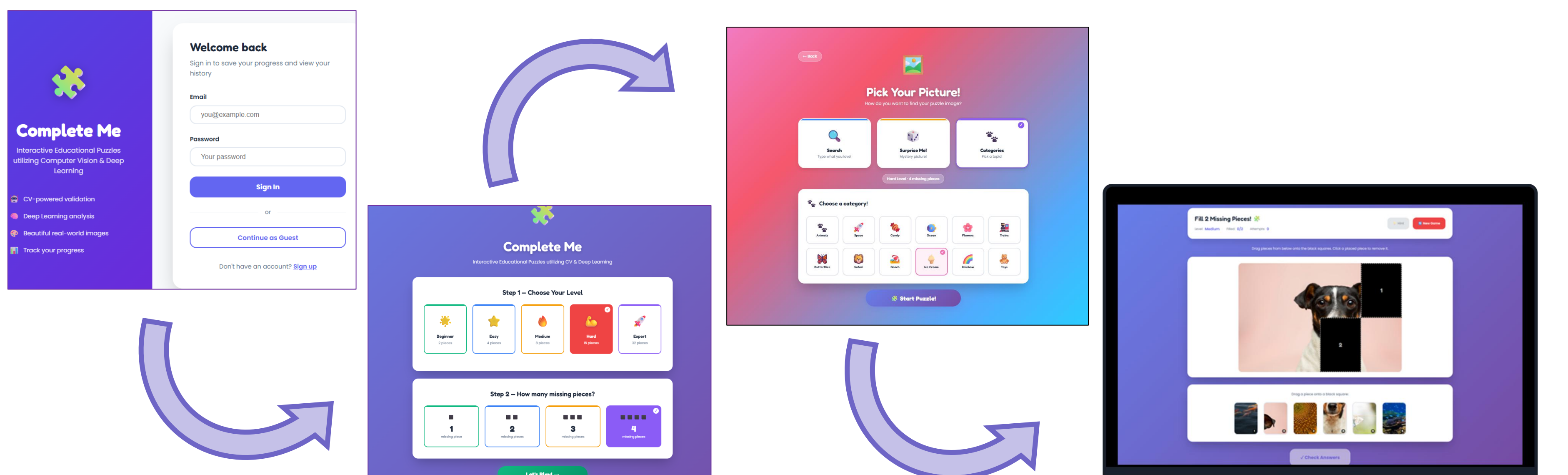
1. Problem description

- Parental Dependency:** Current systems require constant adult supervision to verify whether a child truly understands the image well enough to complete the puzzle correctly.
- Static Content:** Traditional puzzles, both physical and digital, offer a fixed image library with no category selection, causing the learning experience to plateau once the content is exhausted.
- Coordinate-Based Guessing:** Existing digital puzzles validate answers using static (x, y) drop zones, rewarding trial-and-error placement rather than true visual comprehension.



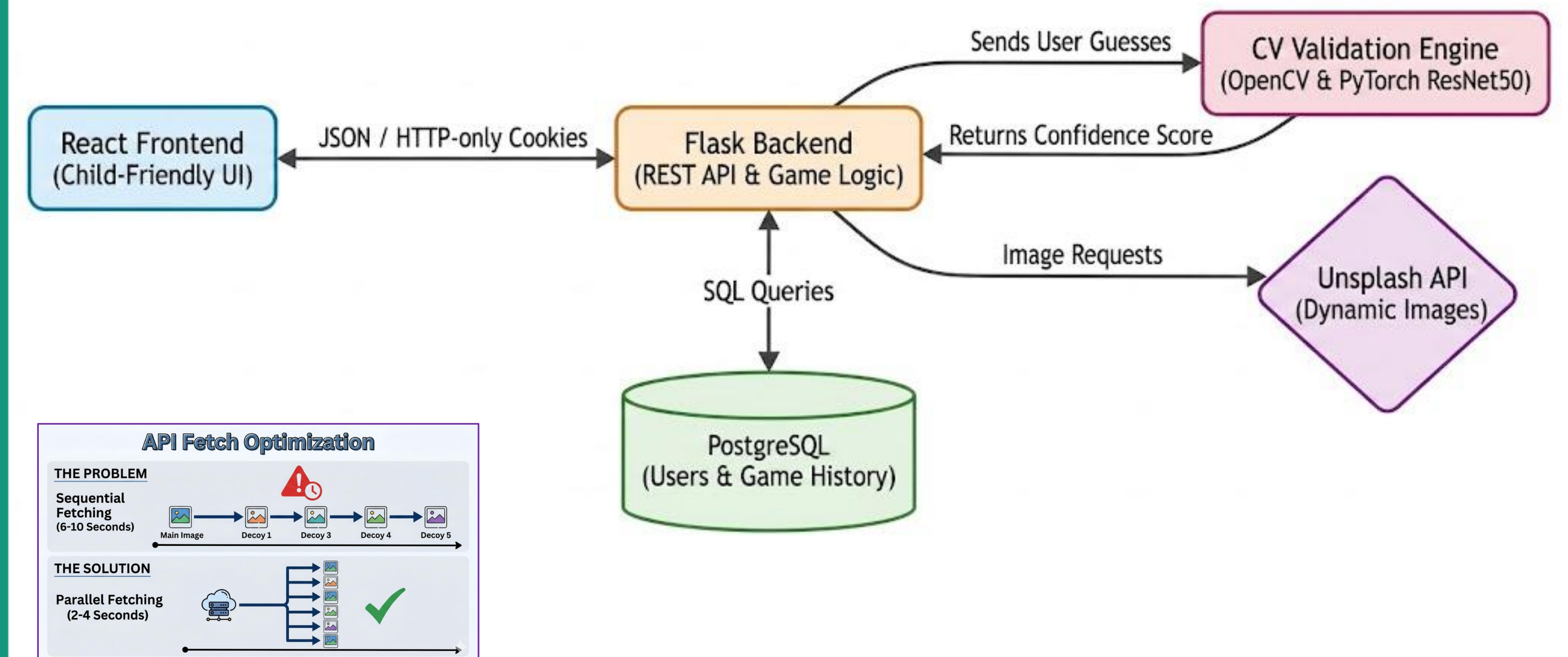
2. The Solution

- Computer Vision Validation:** CompleteMe replaces coordinate-based checks with a multi-layered CV pipeline that analyzes semantic context, color, texture, and edges, mimicking human visual perception.
- Infinite Dynamic Content:** Integration with the Unsplash API generates endless, fresh puzzle scenarios based on safe-listed categories or custom user keywords.
- Independent Learning:** Full login / guest-mode support, combined with immediate feedback, empowers child to learn entirely independently, without adult supervision.



3. Architecture

- Frontend:** React 18 for a responsive, child-friendly UI.
- Backend:** Python/Flask REST API managing game sessions, puzzle state and external API routing.
- Validation Engine:** CV & DL feature extraction and multi-layer image analysis.
- Database:** PostgreSQL for storing user accounts and game progress history.



4. Engineering challenge

Challenge	Description
Puzzle Segmentation	A grid-cutting algorithm splits the image into equal pieces. Missing slots are blacked out for the child to identify and fill.
API Integration	Sequential image fetching caused long load times, solved by fetching all images in parallel using Python threads.
Low-Latency Processing	Real-time DL and CV analysis required careful pipeline optimization to maintain fast response times.
DL-Powered Hint System	Using semantic similarity, ResNet50 finds the closest matching piece and describes it visually, guiding without revealing the answer.
Validation System	CV pipeline ranks all 6 pieces. The user is correct only if their choice ranks first.

